

Cloud Computing Lab — Experiment 3

Network Emulation with Mininet: single and tree Topologies

Field	Details
Name	TODO: Add your name
Roll Number	TODO: Add your roll number
Section	BTECH_CSE_SECB_2023
Course	Cloud Computing Lab
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1. Aim

Install **Mininet** in the Ubuntu VM, verify the installation, and create and test the `single,3` and `tree,3` network topologies using the Mininet CLI commands `links`, `nodes`, `net`, `dump` and `pingall`.

2. Theory

Mininet is a network emulator that creates a realistic virtual network of hosts, switches, controllers and links on a single Linux machine. Each host is a lightweight process with its own network namespace, each link is a virtual Ethernet (veth) pair, and switches are software switches, by default **Open vSwitch (OVS)**. Mininet is widely used to prototype **Software-Defined Networking (SDN)**, where an OpenFlow controller programs the switches centrally. The same idea is used by cloud platforms to build virtual networks (VPCs, virtual switches, overlay networks) on shared physical hardware.

Topology	Command	Structure
minimal (default)	<code>sudo mn</code>	1 switch, 2 hosts
single,N	<code>sudo mn --topo single,N</code>	1 switch with N hosts attached
linear,N	<code>sudo mn --topo linear,N</code>	N switches in a chain, one host per switch
tree,D[,F]	<code>sudo mn --topo tree,D</code>	Tree of switches of depth D and fanout F (default 2), with hosts at the leaves: F^D hosts and $(F^D - 1)/(F - 1)$ switches

For `tree,3` (depth 3, fanout 2) this gives $2^3 = \mathbf{8 \text{ hosts}}$ and $2^3 - 1 = \mathbf{7 \text{ switches}}$.

Useful Mininet CLI commands:

Command	Purpose
<code>nodes</code>	List all nodes (hosts and switches)
<code>links</code>	List all links and their status
<code>net</code>	Show how each node's interfaces are connected
<code>dump</code>	Show each node's type, interfaces, IP address and process ID
<code>pingall</code>	Ping between every pair of hosts and report the drop rate
<code>h1 ping -c 3 h3</code>	Run a command (here <code>ping</code>) on a specific host
<code>exit</code>	Stop the network and leave the CLI

3. Requirements

Item	Value
Host	Windows 11, Oracle VirtualBox 7.2.18
VM	Ubuntu-Cloud-Lab: Ubuntu 26.04.1 LTS, kernel 7.0.0-31-generic, 4 vCPU, 6144 MB RAM, 25 GB disk, NAT
Mininet	2.3.0 (<code>mininet 2.3.0-1.1build1</code>)
Open vSwitch	3.7.1 (<code>openvswitch-switch 3.7.1-2</code>)

Note: the VMs from Experiments 1 and 2 were stored in a folder that was later deleted, so the Ubuntu VM was rebuilt for this experiment. It uses the same configuration as Experiment 1 and was installed from the same checksum-verified Ubuntu 26.04.1 ISO.

4. Procedure

1. **Install Mininet** (see <https://mininet.org/download/> — option 2, "Installation from Packages"):

```
sudo apt update
sudo apt install -y mininet
```

This also installs Open vSwitch (`openvswitch-switch`), which Mininet uses for its switches.

2. **Verify the installation** and run Mininet's built-in test, which builds the minimal topology (h1–s1–h2) and pings between the hosts:

```
sudo mn --version
sudo mn --test pingall
```

3. **Create the single,3 topology** (one switch s1 with hosts h1, h2 and h3) and inspect it from the Mininet CLI:

```
sudo mn --topo single,3
mininet> links
mininet> nodes
mininet> pingall
mininet> net
mininet> dump
mininet> h1 ping -c 3 h3
mininet> exit
```

4. **Create the tree,3 topology** (depth 3, fanout 2) and test it:

```
sudo mn --topo tree,3
mininet> links
mininet> pingall
mininet> exit
```

5. **Clean up** any leftover Mininet state:

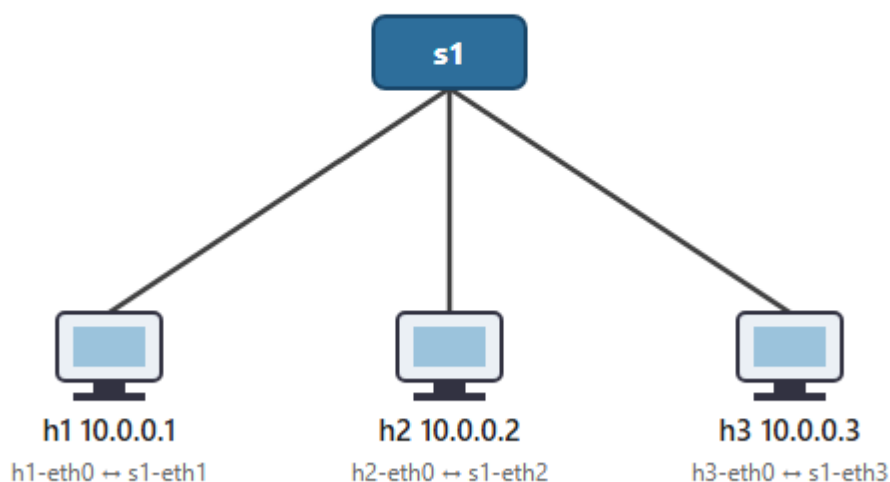
```
sudo mn -c
```

No OpenFlow controller is installed, so Mininet reports *"No default OpenFlow controller found for default switch! Falling back to OVS Bridge"*. The switches then run as standard learning Ethernet bridges, which is enough for these reachability tests. The assignment's reference output shows the same message.

5. Topology Diagrams

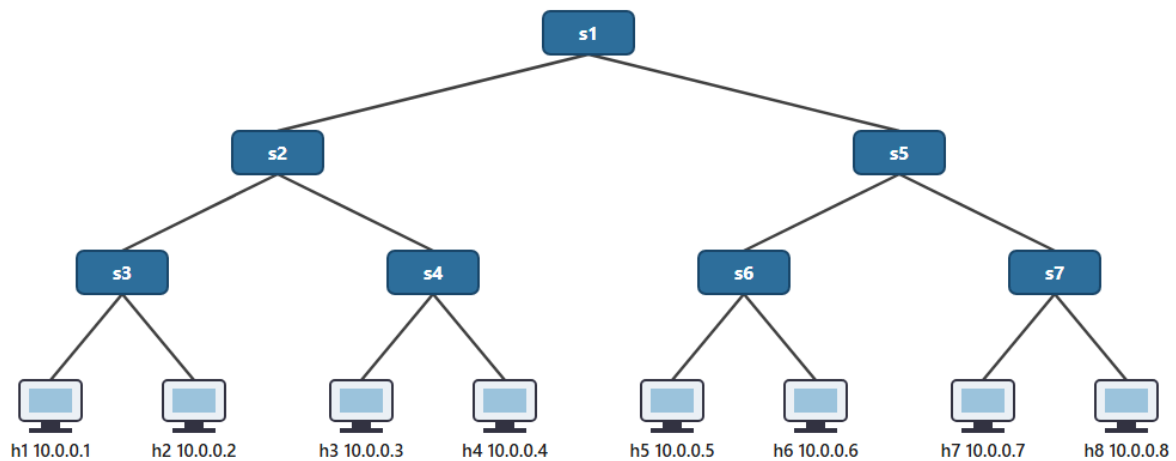
Topology: single,3

`sudo mn --topo single,3` — 1 switch, 3 hosts



Topology: tree,3

sudo mn --topo tree,3 — depth 3, fanout 2: 7 switches, 8 hosts



6. Output

Version and built-in test

```
student@ubuntu-cloud-lab:~$ sudo mn --version
2.3.0
student@ubuntu-cloud-lab:~$ sudo mn --test pingall
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
...
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
...
*** Done
completed in 0.860 seconds
```

single,3

```

student@ubuntu-cloud-lab:~$ sudo mn --topo single,3
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
...
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
h3-eth0<->s1-eth3 (OK OK)
mininet> nodes
available nodes are:
h1 h2 h3 s1
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=12653>
<Host h2: h2-eth0:10.0.0.2 pid=12655>
<Host h3: h3-eth0:10.0.0.3 pid=12657>
<OVSBridge s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=12662>
mininet> h1 ping -c 3 h3
3 packets transmitted, 3 received, 0% packet loss, time 2013ms

```

tree,3

```
student@ubuntu-cloud-lab:~$ sudo mn --topo tree,3
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7
*** Adding links:
(s1, s2) (s1, s5) (s2, s3) (s2, s4) (s3, h1) (s3, h2) (s4, h3) (s4, h4) (s5, s6) (s5, s7) (s6, h
5) (s6, h6) (s7, h7) (s7, h8)
...
*** Starting 7 switches
mininet> links
s1-eth1<->s2-eth3 (OK OK)
s1-eth2<->s5-eth3 (OK OK)
s2-eth1<->s3-eth3 (OK OK)
s2-eth2<->s4-eth3 (OK OK)
s3-eth1<->h1-eth0 (OK OK)
s3-eth2<->h2-eth0 (OK OK)
s4-eth1<->h3-eth0 (OK OK)
s4-eth2<->h4-eth0 (OK OK)
s5-eth1<->s6-eth3 (OK OK)
s5-eth2<->s7-eth3 (OK OK)
s6-eth1<->h5-eth0 (OK OK)
s6-eth2<->h6-eth0 (OK OK)
s7-eth1<->h7-eth0 (OK OK)
s7-eth2<->h8-eth0 (OK OK)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8
h2 -> h1 h3 h4 h5 h6 h7 h8
h3 -> h1 h2 h4 h5 h6 h7 h8
h4 -> h1 h2 h3 h5 h6 h7 h8
h5 -> h1 h2 h3 h4 h6 h7 h8
h6 -> h1 h2 h3 h4 h5 h7 h8
h7 -> h1 h2 h3 h4 h5 h6 h8
h8 -> h1 h2 h3 h4 h5 h6 h7
*** Results: 0% dropped (56/56 received)
```

7. Screenshots

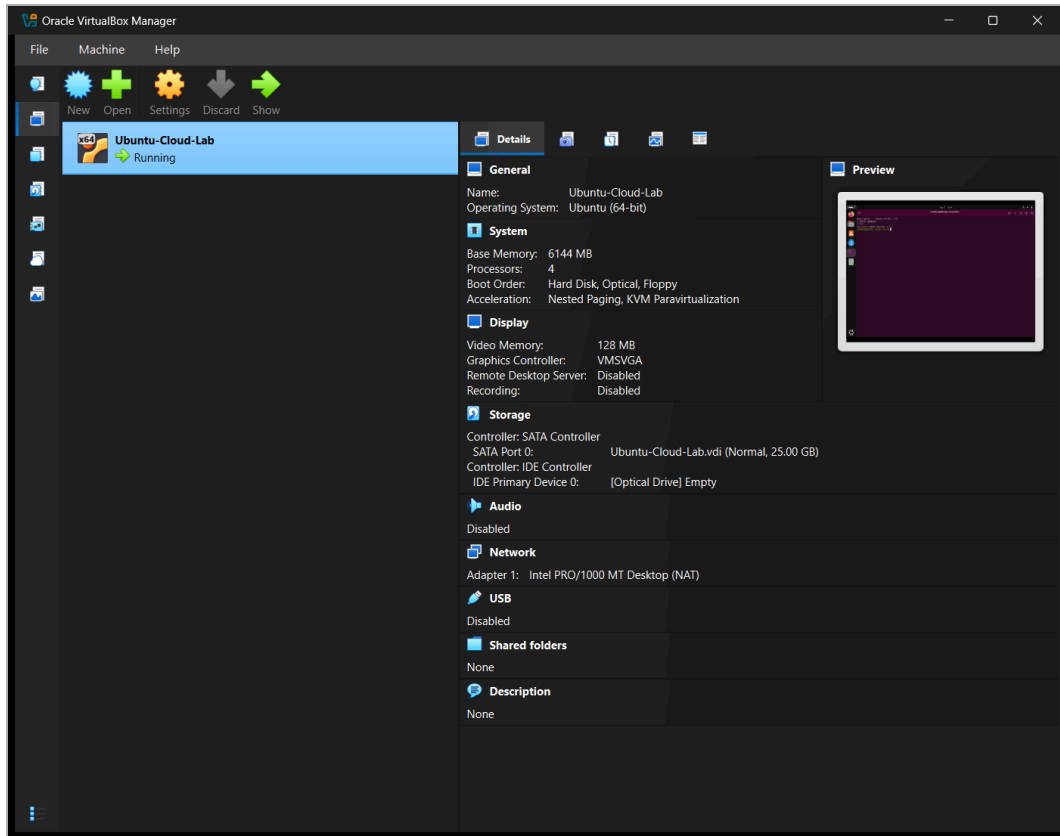


Figure 0: Ubuntu VM running in Oracle VirtualBox

```

Sep 21 13:42
student@ubuntu-cloud-lab: ~
Preparing to unpack .../11-mininet_2.3.0-1.1build1_amd64.deb...
Unpacking mininet (2.3.0-1.1build1)...
Selecting previously unselected package openvswitch-common.
Preparing to unpack .../12-openvswitch-common_3.7.1-2_amd64.deb...
Unpacking openvswitch-common (3.7.1-2)...
Selecting previously unselected package openvswitch-switch.
Preparing to unpack .../13-openvswitch-switch_3.7.1-2_amd64.deb...
Unpacking openvswitch-switch (3.7.1-2)...
Selecting previously unselected package python3-openvswitch.
Preparing to unpack .../14-python3-openvswitch_3.7.1-2_amd64.deb...
Unpacking python3-openvswitch (3.7.1-2)...
Setting up net-tools (2.10-2ubuntu1)...
Setting up libcgroup3:amd64 (3.1.0-2build1)...
Setting up libevent-2.1-7t64:amd64 (2.1.12-stable-10ubuntu0.1)...
Setting up cgroup-tools (3.1.0-2build1)...
Setting up python3-sortedcontainers (2.4.0-2build1)...
Setting up libxdp1:amd64 (1.6.2-1ubuntu1)...
Setting up python3-packaging (26.0-1)...
Setting up libunbound8:amd64 (1.24.2-1ubuntu2.2)...
Setting up iperf (2.2.1+dfsg-1build1)...
Setting up socat (1.8.1.1-1ubuntu0.1)...
Setting up python3-netifaces:amd64 (0.11.0-2build7)...
Setting up openvswitch-common (3.7.1-2)...
Setting up mininet (2.3.0-1.1build1)...
Setting up openvswitch-switch (3.7.1-2)...
update-alternatives: using /usr/lib/openvswitch-switch/ovs-vswitchd to provide /usr/sbin/ovs-vswitchd (ovs-vswitchd) in auto mode
Created symlink '/etc/systemd/system/openvswitch-switch.service.requires/ovs-record-hostname.service' -> '/usr/lib/systemd/system/ovs-record-hostname.service'.
Created symlink '/etc/systemd/system/openvswitch.service' -> '/usr/lib/systemd/system/openvswitch-switch.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/openvswitch-switch.service' -> '/usr/lib/systemd/system/openvswitch-switch.service'.
Setting up python3-openvswitch (3.7.1-2)...
Processing triggers for libc-bin (2.43-2ubuntu2.4)...
Processing triggers for ufw (0.36.2-9build1)...
Processing triggers for man-db (2.13.1-1build1)...
student@ubuntu-cloud-lab:~$

```

Figure 1: Installing Mininet and Open vSwitch with `sudo apt install -y mininet`

```
Sep 21 13:43
student@ubuntu-cloud-lab: ~
student@ubuntu-cloud-lab:~$ sudo mn --version
2.3.0
student@ubuntu-cloud-lab:~$ sudo mn --test pingall
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller

*** Starting 1 switches
s1 ...
*** Waiting for switches to connect
s1
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
*** Stopping 0 controllers

*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
completed in 0.860 seconds
student@ubuntu-cloud-lab:~$
```

Figure 2: `sudo mn --version` and `sudo mn --test pingall` (0% dropped, 2/2)

```
Sep 21 13:43
student@ubuntu-cloud-lab: ~ -- sudo mn --topo single,3
student@ubuntu-cloud-lab:~$ sudo mn --topo single,3
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller

*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
h3-eth0<->s1-eth3 (OK OK)
mininet> nodes
available nodes are:
h1 h2 h3 s1
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet>
```

Figure 3: `sudo mn --topo single,3` with `links`, `nodes`, `pingall` (0% dropped, 6/6)

```
Sep 21 13:44
student@ubuntu-cloud-lab: ~ — sudo mn --topo single,3

mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=12653>
<Host h2: h2-eth0:10.0.0.2 pid=12655>
<Host h3: h3-eth0:10.0.0.3 pid=12657>
<OVSBridge s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=12662>
mininet> h1 ping -c 3 h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=1.95 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=0.040 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=0.042 ms

--- 10.0.0.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
rtt min/avg/max/mdev = 0.040/0.676/1.948/0.898 ms
mininet>
```

Figure 4: single,3: net, dump, h1 ping -c 3 h3

```
Sep 21 13:46
student@ubuntu-cloud-lab: ~ — sudo mn --topo tree,3

student@ubuntu-cloud-lab:~$ sudo mn --topo tree,3
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7
*** Adding links:
(s1, s2) (s1, s5) (s2, s3) (s2, s4) (s3, h1) (s3, h2) (s4, h3) (s4, h4) (s5, s6) (s5, s7) (s6, h5) (s6, h6) (s7, h7) (s7, h8)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8
*** Starting controller

*** Starting 7 switches
s1 s2 s3 s4 s5 s6 s7 ...
*** Starting CLI:
mininet> links
s1-eth1->s2-eth3 (OK OK)
s1-eth2->s5-eth3 (OK OK)
s2-eth1->s3-eth3 (OK OK)
s2-eth2->s4-eth3 (OK OK)
s3-eth1->h1-eth0 (OK OK)
s3-eth2->h2-eth0 (OK OK)
s4-eth1->h3-eth0 (OK OK)
s4-eth2->h4-eth0 (OK OK)
s5-eth1->s6-eth3 (OK OK)
s5-eth2->s7-eth3 (OK OK)
s6-eth1->h5-eth0 (OK OK)
s6-eth2->h6-eth0 (OK OK)
s7-eth1->h7-eth0 (OK OK)
s7-eth2->h8-eth0 (OK OK)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8
h2 -> h1 h3 h4 h5 h6 h7 h8
h3 -> h1 h2 h4 h5 h6 h7 h8
h4 -> h1 h2 h3 h5 h6 h7 h8
h5 -> h1 h2 h3 h4 h6 h7 h8
h6 -> h1 h2 h3 h4 h5 h7 h8
h7 -> h1 h2 h3 h4 h5 h6 h8
h8 -> h1 h2 h3 h4 h5 h6 h7
*** Results: 0% dropped (56/56 received)
mininet>
```

Figure 5: sudo mn --topo tree,3 with Links and pingall (0% dropped, 56/56)

8. Result

Mininet 2.3.0 was installed in the Ubuntu 26.04.1 VM along with Open vSwitch 3.7.1. The built-in test passed (2/2 received). The `single,3` topology (1 switch, 3 hosts, 3 links) and the `tree,3` topology (7 switches, 8 hosts, 14 links) were created, all links reported (OK OK), and `pingall` reported **0% dropped** in both cases: 6/6 packets for `single,3` and 56/56 packets for `tree,3`.

9. Conclusion

Mininet lets a complete network of hosts, switches and links be emulated inside one virtual machine, using Linux network namespaces for hosts, veth pairs for links and Open vSwitch for switches. Changing a single command-line option (`--topo single,3` or `--topo tree,3`) builds a different topology in seconds, and the CLI commands `links`, `nodes`, `net`, `dump` and `pingall` make it easy to inspect and verify the network. This software-defined, reproducible way of building networks is the same principle cloud providers use for the virtual networks behind their VMs and containers.